

To what extent does the River Modau conform to the Bradshaw Model?



Candidate Code:

Word count:

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1 Introduction

1.1 Bradshaw Model

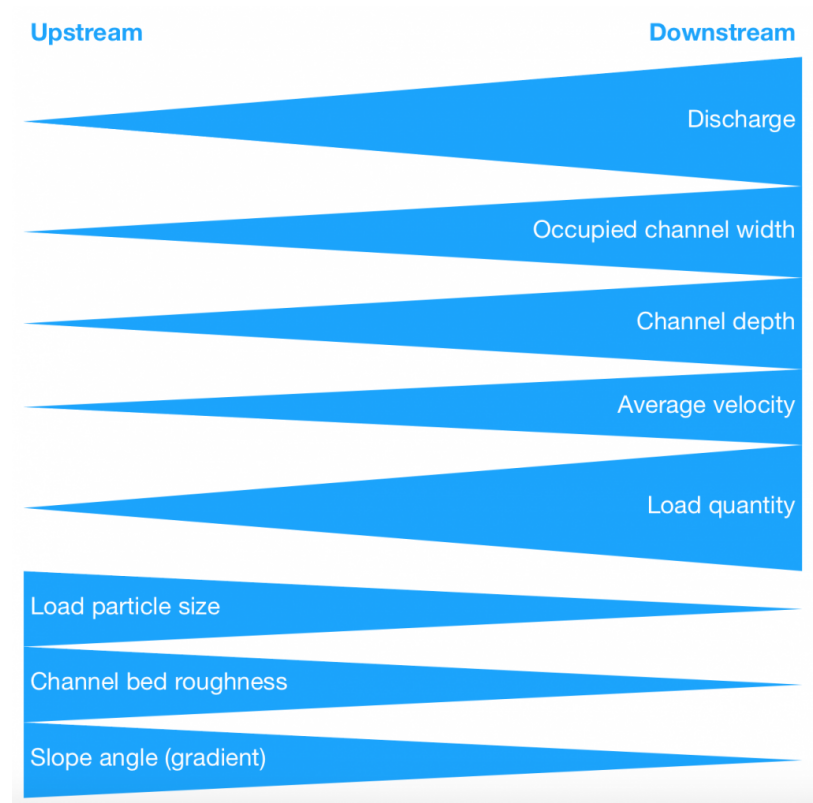


Figure 1: Bradshaw Model (Internet Geography)

The Bradshaw Model is a geographical model which theorizes the characteristics of a river, stream, or tributary as one moves downstream from the source to the mouth (En-Academic). However, the model can be read from both ways, upstream and downstream. Here it portrays how river discharge, channel width, depth, average velocity, and load quantity increase as one proceeds downstream. On the other hand, moving down stream, the load particle size, channel bed roughness, and gradient decrease (Cool Geography). Despite showing the predicted river characteristics moving downstream, the Bradshaw Model does not encompass all the features of the river. Properties such as channel efficiency, river valley shapes and river landforms are exempt from this paradigm.

1.2 Aim

The aim of this following research is to determine the accuracy of the Bradshaw Model in predicting the characteristics of a river as it flows from its source down to its mouth. In modern times, many anthropogenic changes to natural landforms have been made, either to accommodate growing settlements or to make way for various projects at an ideal, specific location. To evaluate the impact of these modifications on, and whether all rivers do act in accordance to, the Bradshaw model, the following question, “*To what extent does the River Modau conform to the Bradshaw Model?*”, was formulated.

The survey for this study was conducted on the 26th of June, 2025 between the hours 8 a.m. to 4 p.m.. This investigation adheres to Option A: Freshwater, specifically **Drainage Basin Hydrology** and **Geomorphology** (Clastify).

1.3 Modau River

The Modau River is a small river located in the southern part of the state of Hesse, Germany. Geographically, it originates in the Odenwald hills near Neunkirchen and flows northwest through a series of valleys and lowlands.

The river stretches approximately 44 kilometres before it empties into the Rhine River near Stockstadt am Rhine. As it flows through town like Ober-Ramstadt, Pfungstadt, and parts of Darmstadt, the Modau shapes the surrounding landscape by contributing to the local drainage system and influencing soil fertility.

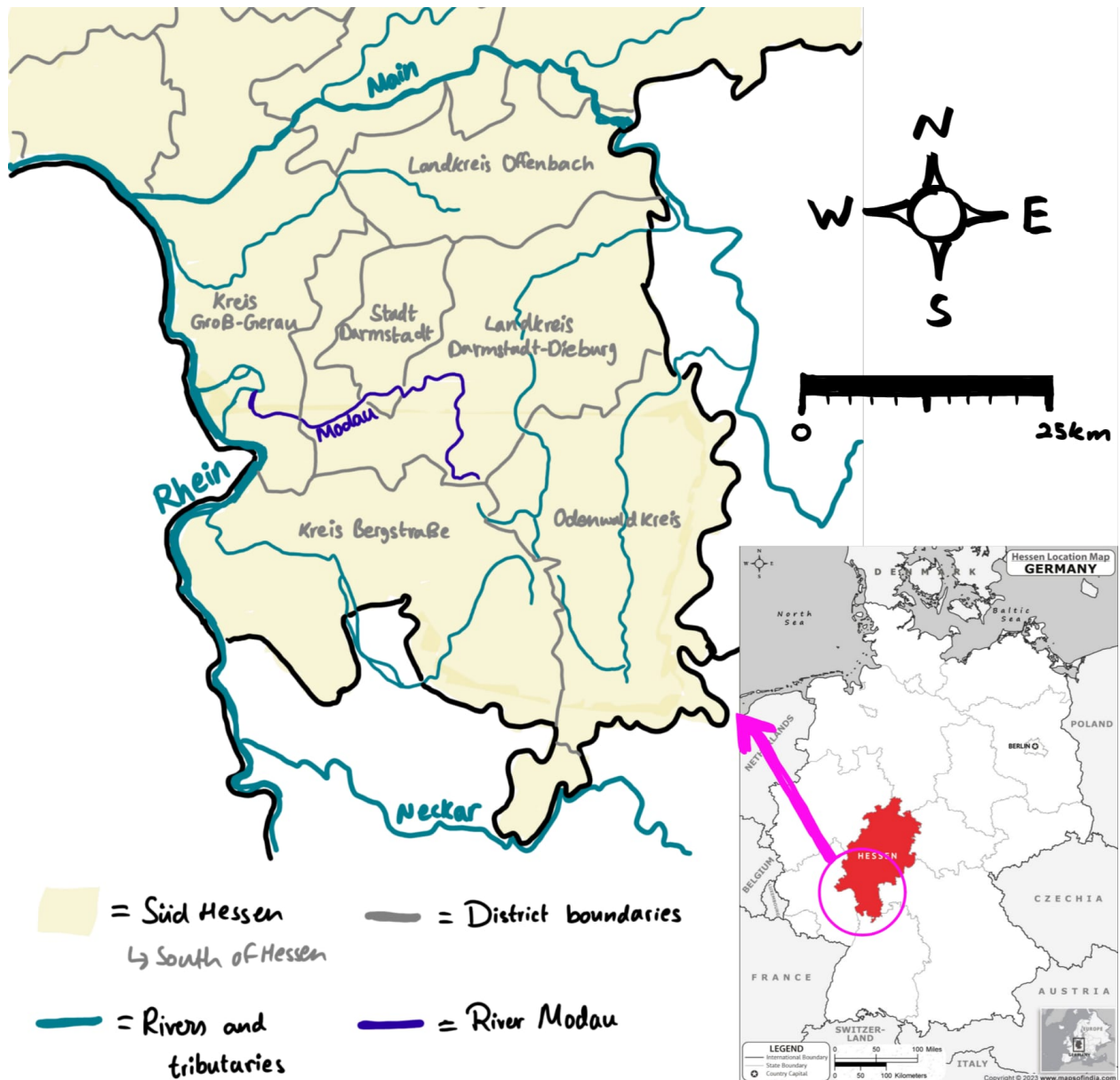


Figure 2: River Modau geographical location in relation to Map of Germany (MapsofIndia)

Its drainage basin lies within a temperate climate zone, and the river plays a role in regional hydrology, especially in terms of groundwater surface-water runoff.

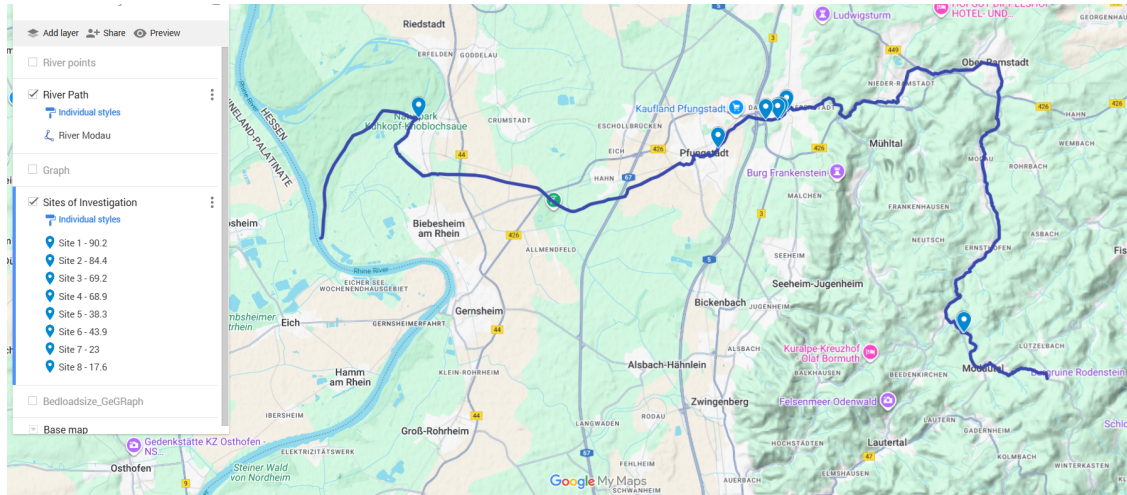


Figure 3: Map of Modau and the sites of investigation created using Google Maps.

As shown in the map, one can clearly follow the tributary from its source in the hills to the east, its upper course till Ober-Ramstadt where the river path is straight and in a v-shape valley, its middle course till Pfungstadt where it has entered its flat-floored valley leaving behind the hills with slight meanders which are more prominent when zoomed, and finally, the lower course as it joins the Rhine River. Here it is clearly a flat valley as the beige colour represents the surrounding farmland by the river. Meanders are clearer and more visible. As it flows to the River Rhine, the Modau has tributaries that flow into it the more we descend in altitude.

2 Hypotheses

Hypothesis 1: If one moves downstream, then the **velocity**¹ increases as **channel efficiency** increases.

This is as channel efficiency is entirely dependent on the **hydraulic radius**.² An increase in hydraulic radius leads to an increase in channel efficiency, which means the proportion of water in contact with the river banks and bed is less, meaning there is less drag, allowing the velocity to increase.

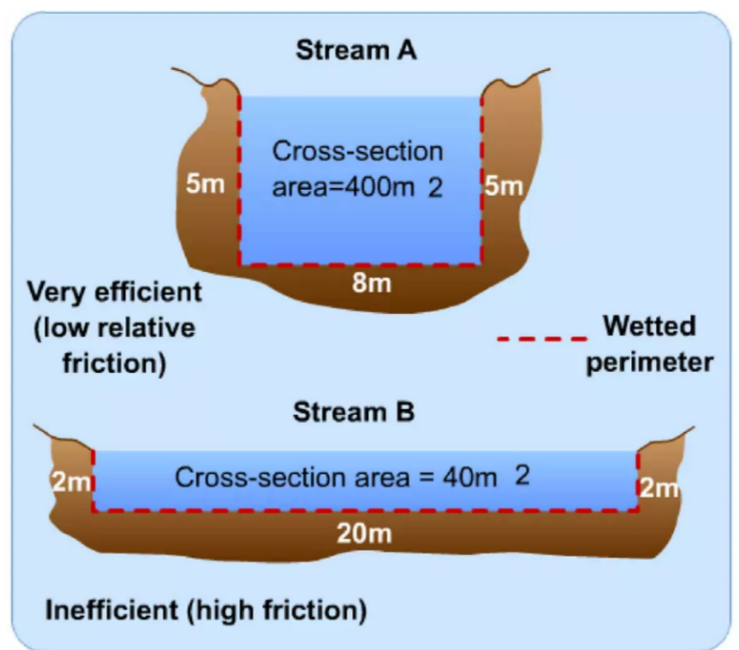


Figure 4: Channel Efficiency example (Slideshare).

As portrayed above, a semi-circular channel shape provides less contact with the river banks and bed compared to a flat floored channel, meaning the semi-circular channel has a higher channel efficiency. Furthermore, there is more vertical and lateral erosion as one moves downstream, increasing channel efficiency as the channel walls get smoother, causing less friction.

¹The speed at which the river flows measured in either m s^{-1} or km s^{-1} .

²The cross-sectional area divided by the wetted perimeter provides the hydraulic radius.

Hypothesis 2: If one moves downstream, then there is an increase in **river discharge**³, because **channel width**⁴ and **channel depth**⁵ increase.

This occurs as the more downstream one goes, the more the water erodes the banks and the river bed through abrasion due to its sediments, increasing channel depth and width. This increases the carrying capacity of the river, allowing for a greater river discharge.

Hypothesis 3: Bedload⁶ **size** will decrease as we go downstream, as there are more erosional processes taking place due to the increased amount of sediment being carried in the river.

This is reasonable as due to an increased amount of load carried by the river as it moves downstream, there are more erosional processes taking place and to a greater extent. These include processes like traction, corrosion, abrasion, and more where sediment drags along the river bed eroding itself and the bed of the river, acidic conditions in the water can cause sediments like limestone to erode, small rocks hitting each other due to the river current causing them to wear each other down, and more.

³The volume of water flowing through a channel at a certain point of time, measured in cumecs ($\text{m}^3 \text{s}^{-1}$).

⁴How wide the river is, measured laterally, in meters, at the surface from river bank to river bank where the water touches the land.

⁵How deep the river bed is from the surface of the water, measured at set intervals perpendicular to river flow in order to account for the irregular shape of the river bed, measured in meters.

⁶The sediment transported along the bottom of the river due to the river flow (Fiveable).

3 Methodology

3.1 Velocity (Digital)

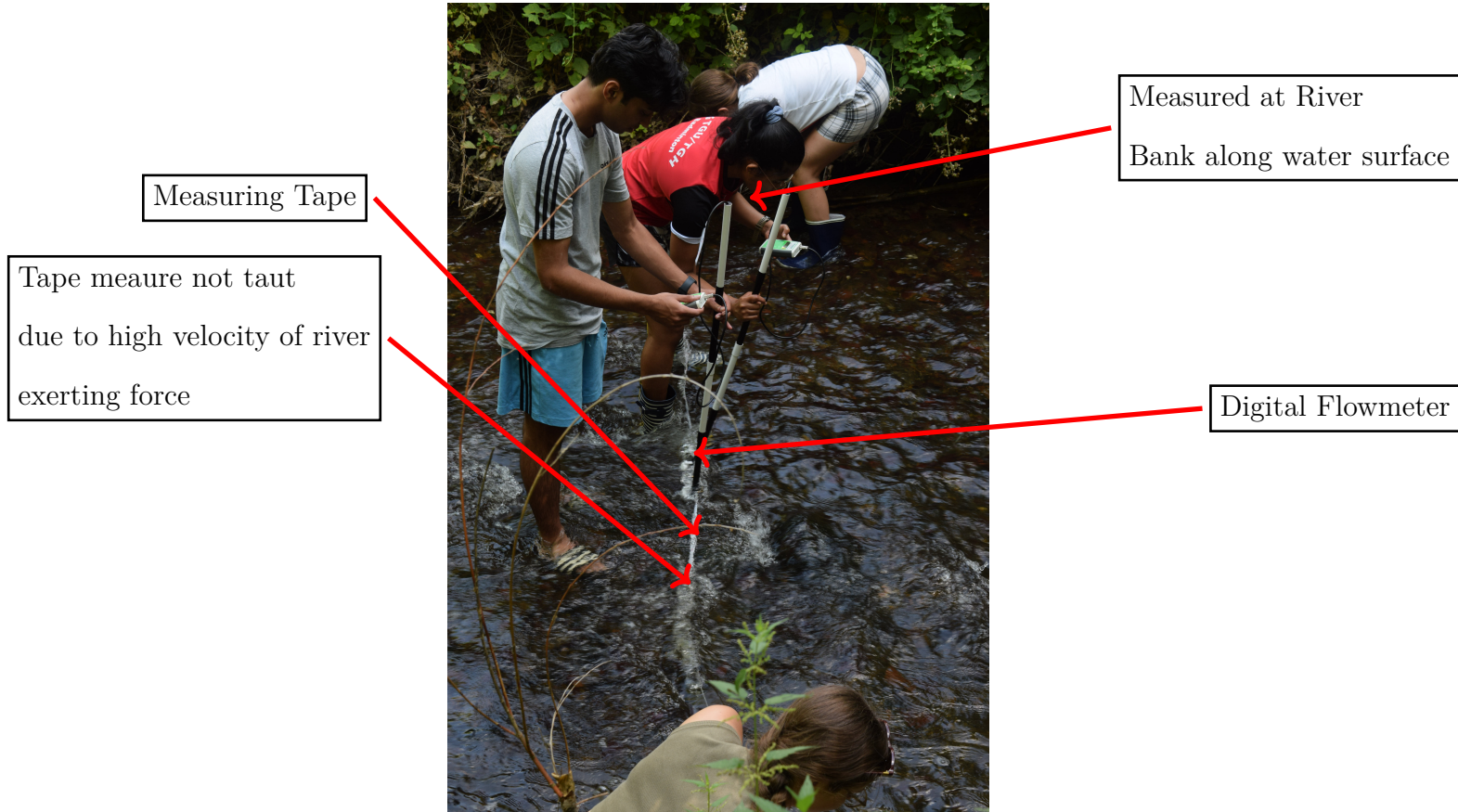


Figure 5: Digital Velocity measured at Site 3.

Velocity was measured in two ways for the following investigation, digital and analogue. To measure digitally, first the width of the river channel was measured at each site and the middle of the channel was recorded. This middle point in the river was where the flow-meter was placed at the surface, with the propeller facing against the flow, in order to measure the velocity as a form of **purposive sampling**⁷.

⁷**Purposive sampling** is when a sample is chosen based on a predetermined feature, which is beneficial to the outcome of the study.

Velocity Equation when digital flow-meter is not used:

$$v = \frac{\Delta s}{\Delta t}$$

v = velocity (m/s)
 Δs = change in distance (m)
 Δt = change in time (s)

Purposive sampling is used, as the least drag is present in the middle of the river channel and away from the river bed, which is a pre-defined characteristic chosen for higher accuracy. This was measured 4 times as the data kept fluctuating by one decimal point, then the average was calculated. A digital flow-meter was used for the data as it is more accurate than an analogue measurements as obstacles in the river, can halt or alter the flow of the float.

3.2 Channel Width



Figure 6: Measurement of Channel width in middle course.

Measurement of the channel width was taken by using a tape measure. The tape measure was placed perpendicular to river flow and the measurement was taken from river bank to river bank at the points where the water meets the bank. During this the tape measure was pulled taut across the water surface before the measurements were taken. There were 3 measurements made for width and the sampling technique used was pragmatic sampling.

This method of sampling was used as the point in the sites chosen to measure width were chosen as they were the most accessible with the least amount of obstructions, which would cause uncertainties.

3.3 Channel Depth

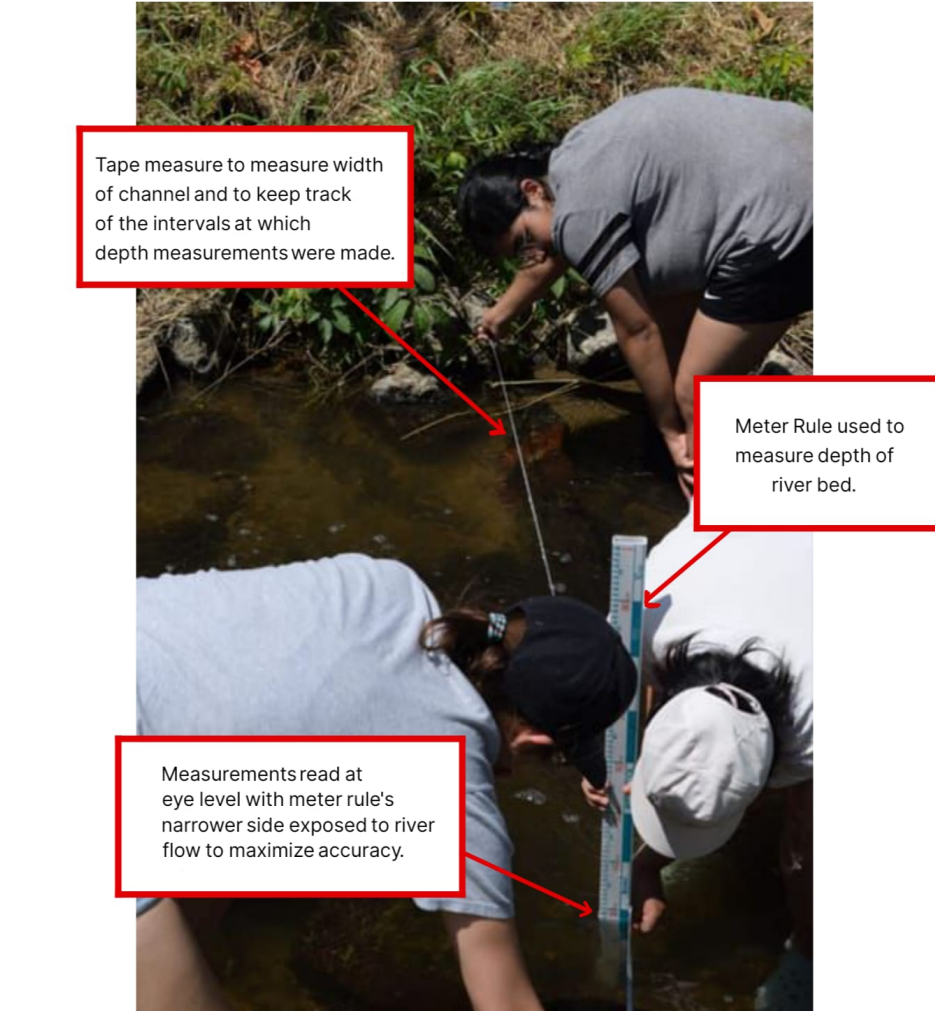


Figure 7: Depth measurements taken at river

Measurement of the channel depth was taken through the use of a meter rule. A tape measure spanning the width of the river was placed in order to provide the overall width, which would then help determine the intervals at which to measure the depth across the river channel. The meter rule was placed with its narrower side exposed to river flow and the distance between the water surface and the riverbed was measured and recorded.

Pragmatic sampling was also utilised here as places that had fewer significant protrusions in the river bed, such as large rocks, were avoided to allow for a more accurate measurement of the river bed. These measurements were taken 3 times at each interval and had the average

calculated for each of the intervals. Then for the overall depth, the mean of the averages was used.

3.4 Cross-Sectional Area

Cross-sectional area is measured by multiplying the mean width with the mean depth.

$$A = \text{Mean Width} \times \text{Mean Depth}$$

$$A_{\text{Site 1}} = 2.02 \times 0.145 = 0.2929 \text{ m}^2$$

3.5 Channel Efficiency

1. Channel efficiency = Hydraulic radius $\rightarrow$ The higher the radius, the greater the efficiency.
2. Cross-Sectional Area $\div$ Wetted Perimeter = Hydraulic Radius

$$R = \frac{A}{P}$$

$$R_{\text{Site 1}} = \frac{0.2929}{2.03} = 0.144 \text{ m (3 s.f.)}$$

3.6 River Discharge

$$\text{Discharge (cumecs)} = \text{Cross-Sectional Area (m}^2\text{)} \times \text{Velocity (m s}^{-1}\text{)}$$

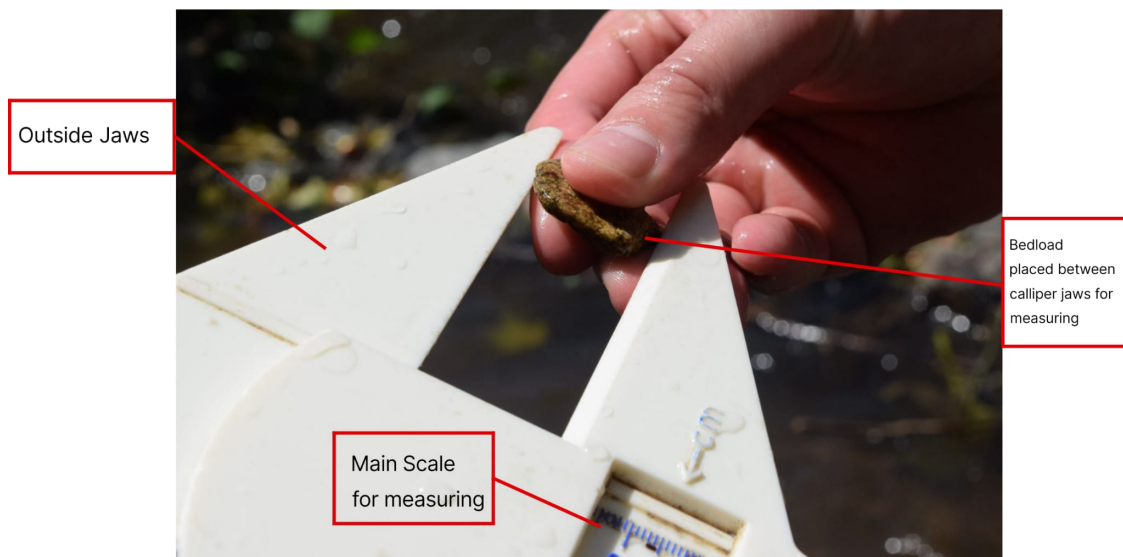
The mean depth and mean width are first used to measure cross-sectional area (m²), then the

velocity is measured through either a digital flow-meter or with a stopwatch, a cork, measuring tape and ranging poles ($\times 4$). Then the average velocity is taken and multiplied by the area to get the discharge for the river.

$$Q = A \times v$$

$$Q_{\text{Site 1}} = 0.2929 \times 0.5725 = 0.168(3s.f.) \text{ m}^3/\text{s}$$

3.7 Bedload Size



To measure bedload size, a calliper was used and measurements on the chosen rock's length, width and breadth were taken. These results were then averaged and recorded on a table. 10 samples were measure at every site, the averages of the average sizes of these samples were taken in order to determine the average bedload size at each site.

Systematic sampling was used in this case so as to broaden the variety of bedload samples used to be measured in order to provide a higher level of accuracy.

4 Processed Data

4.1 First Hypothesis

4.1.1 Velocity

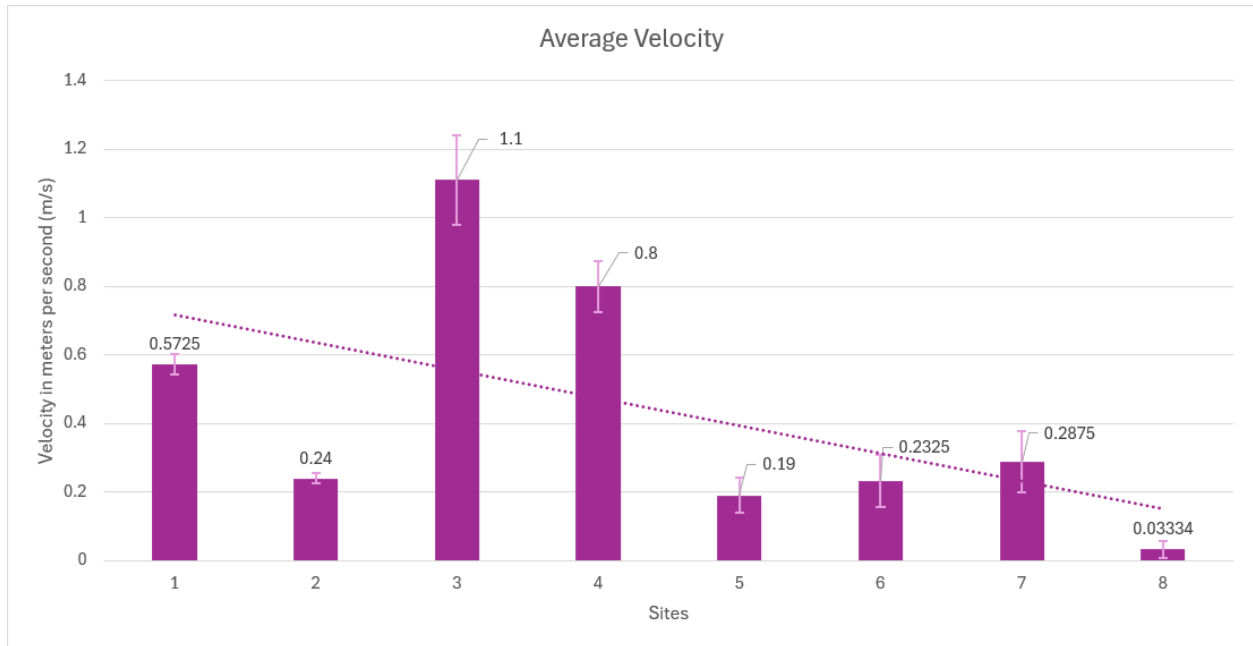


Figure 8: Average Velocity at Each Site

4.1.2 Channel Efficiency

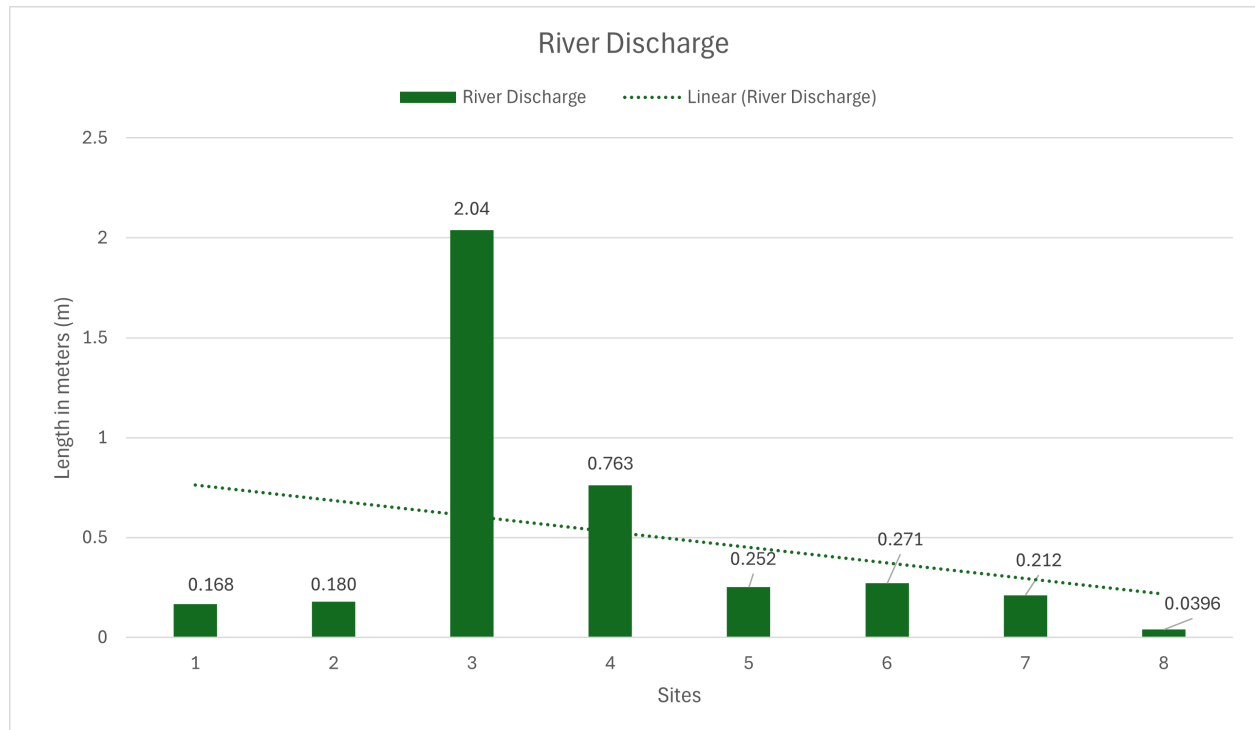


Figure 9: Channel Efficiency at Different Sites

4.2 Second Hypothesis

4.2.1 Channel Width

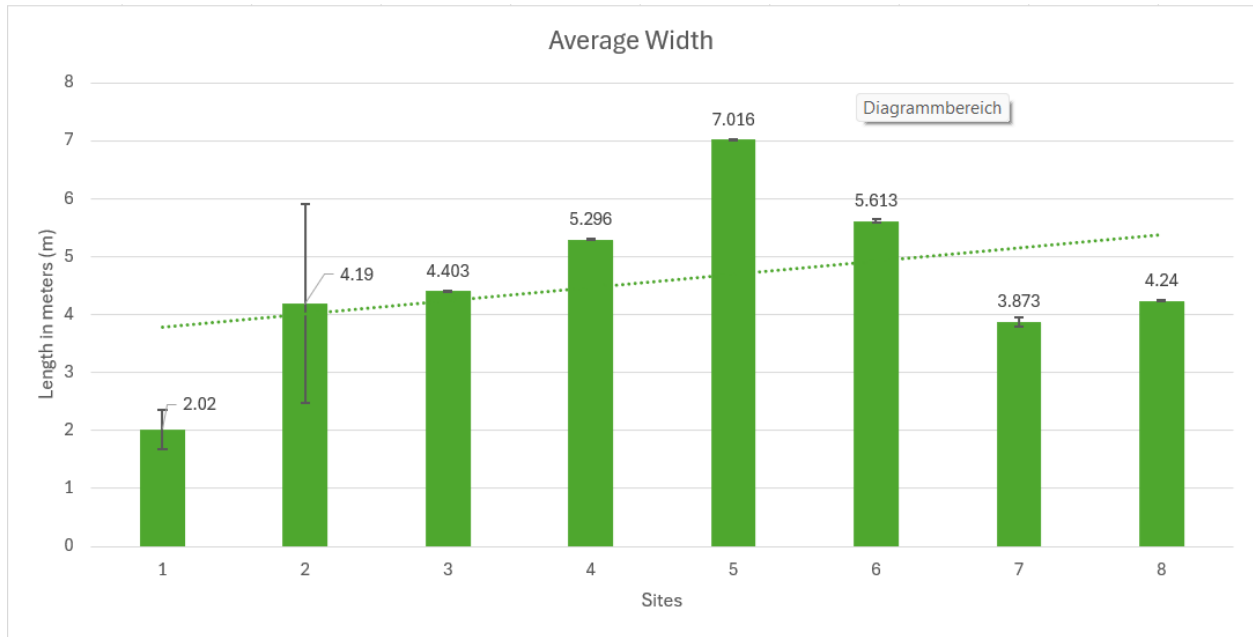


Figure 10: Average Channel Width at Each Site

4.2.2 Channel Depth

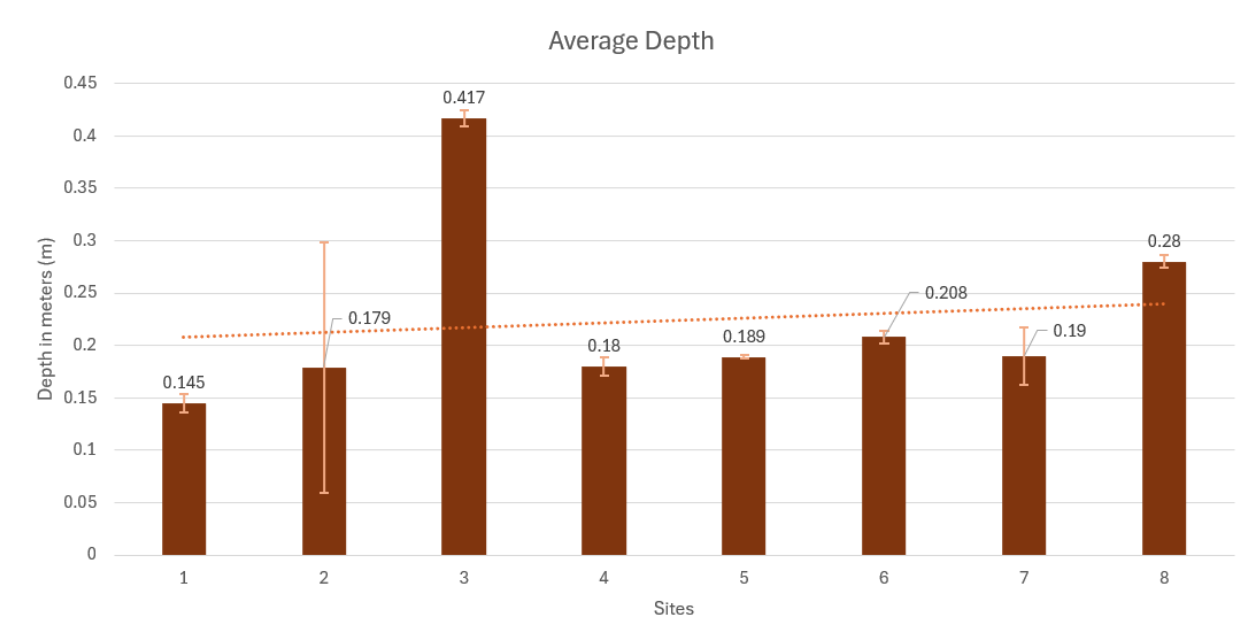


Figure 11: Average Channel Depth at Each Site

4.2.3 Discharge

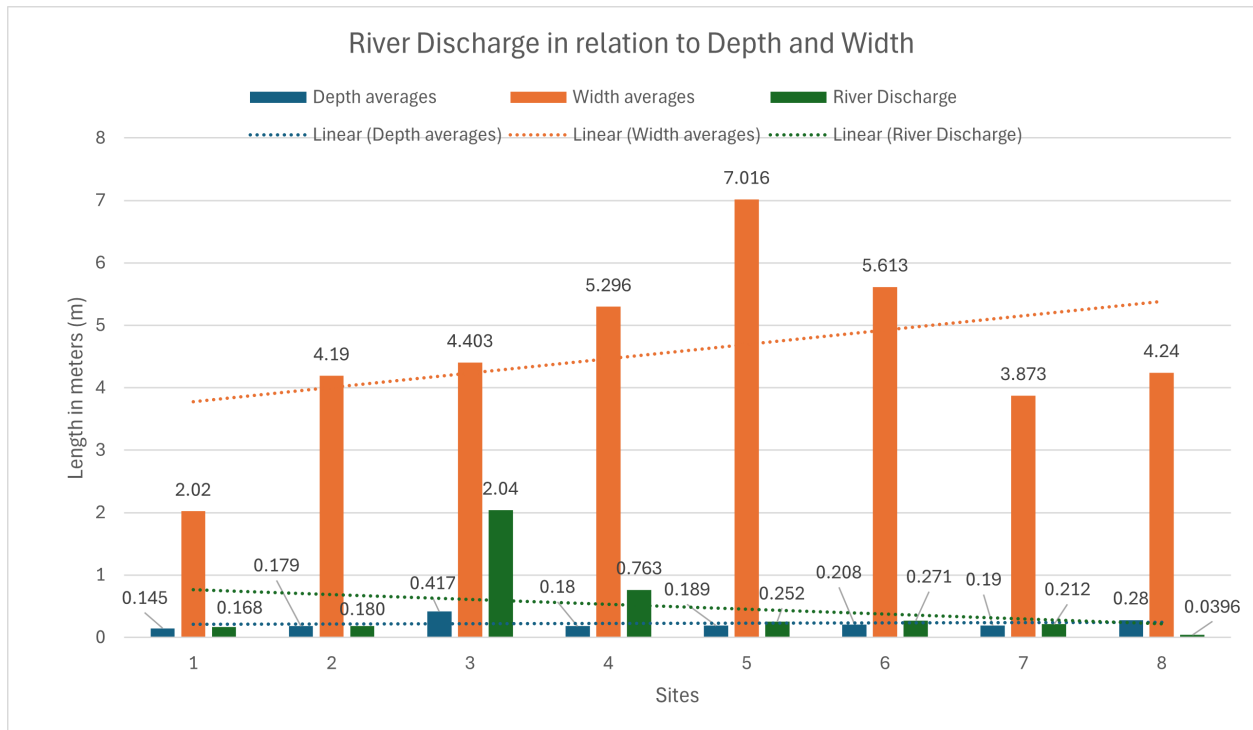


Figure 12: Graph of River Discharge in relation to Depth and Width of the river channel at each site.

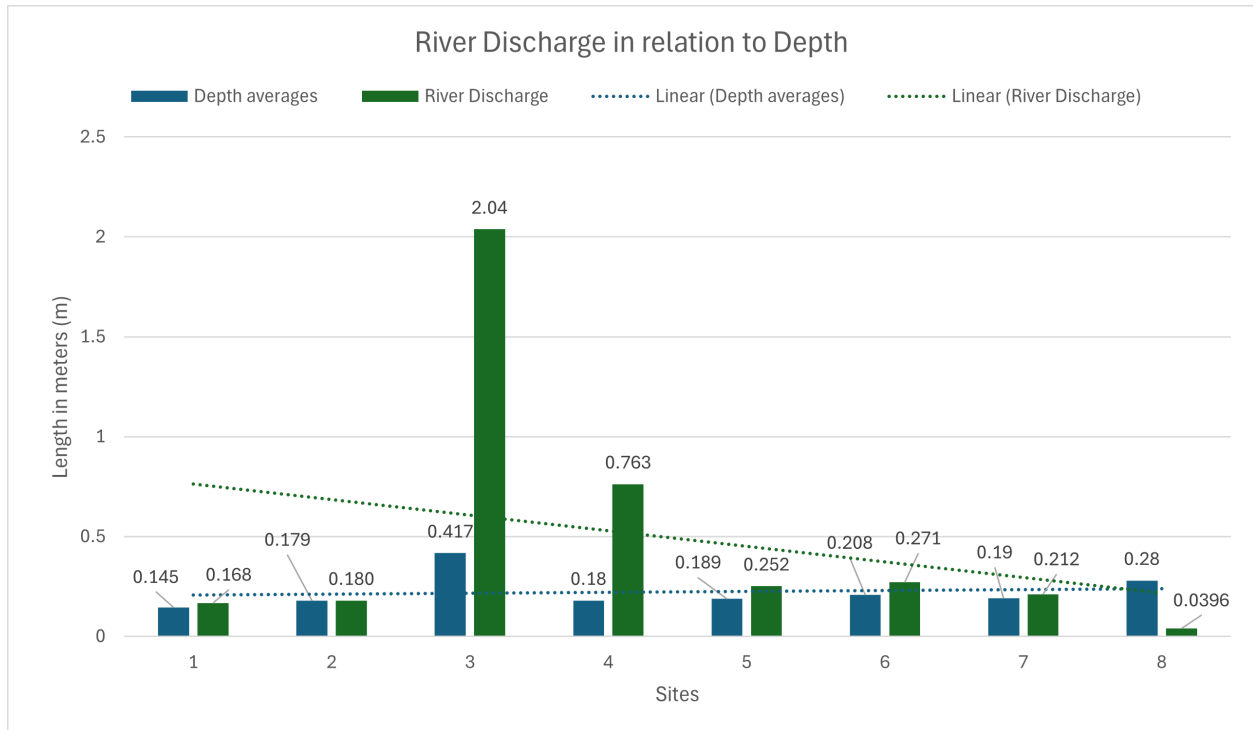


Figure 13: Average Channel Depth at Each Site

4.3 Wetted Perimeter

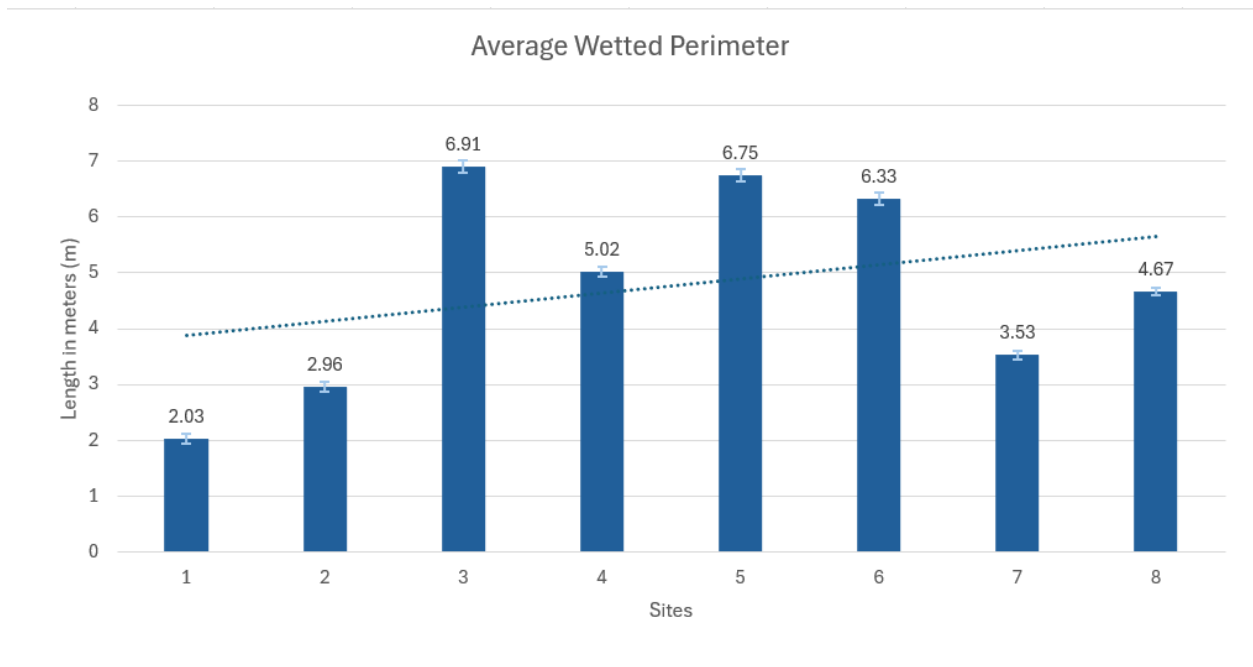


Figure 14: Wetted Perimeter at Different Sites

4.4 Cross-Sectional Area

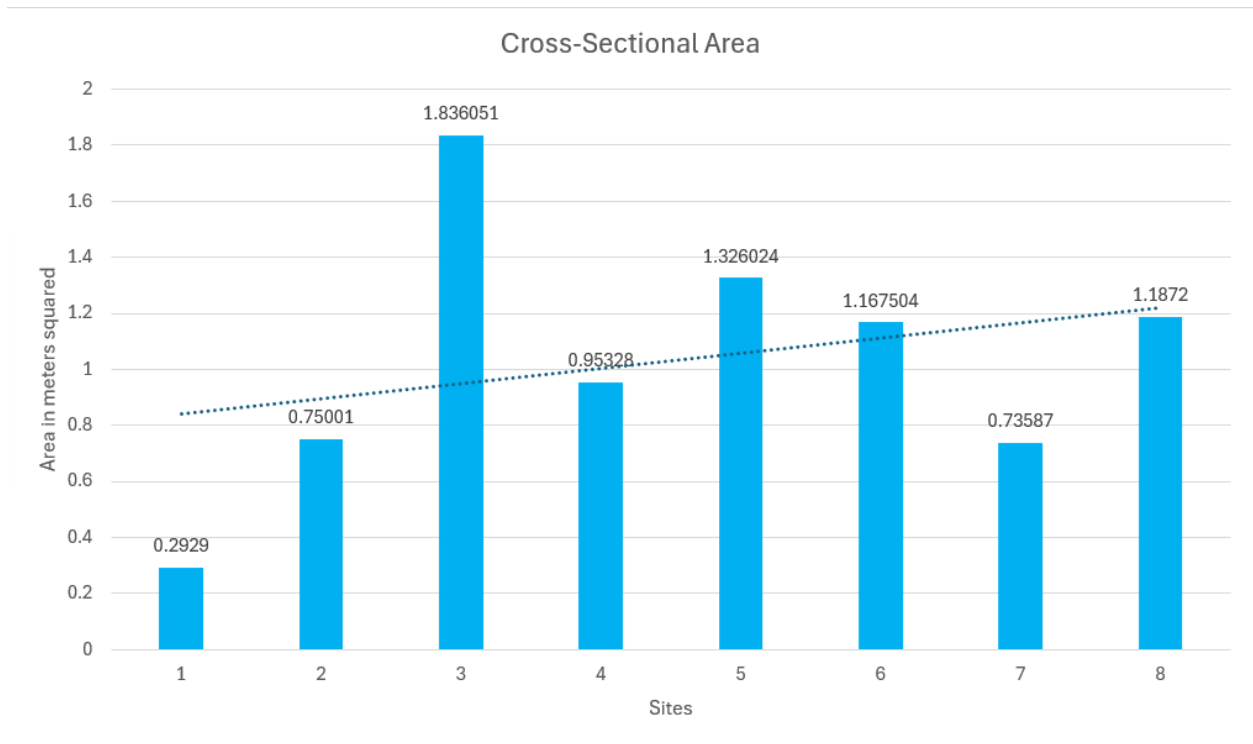


Figure 15: Cross-Sectional Area at Different Sites

4.5 Hypothesis 3

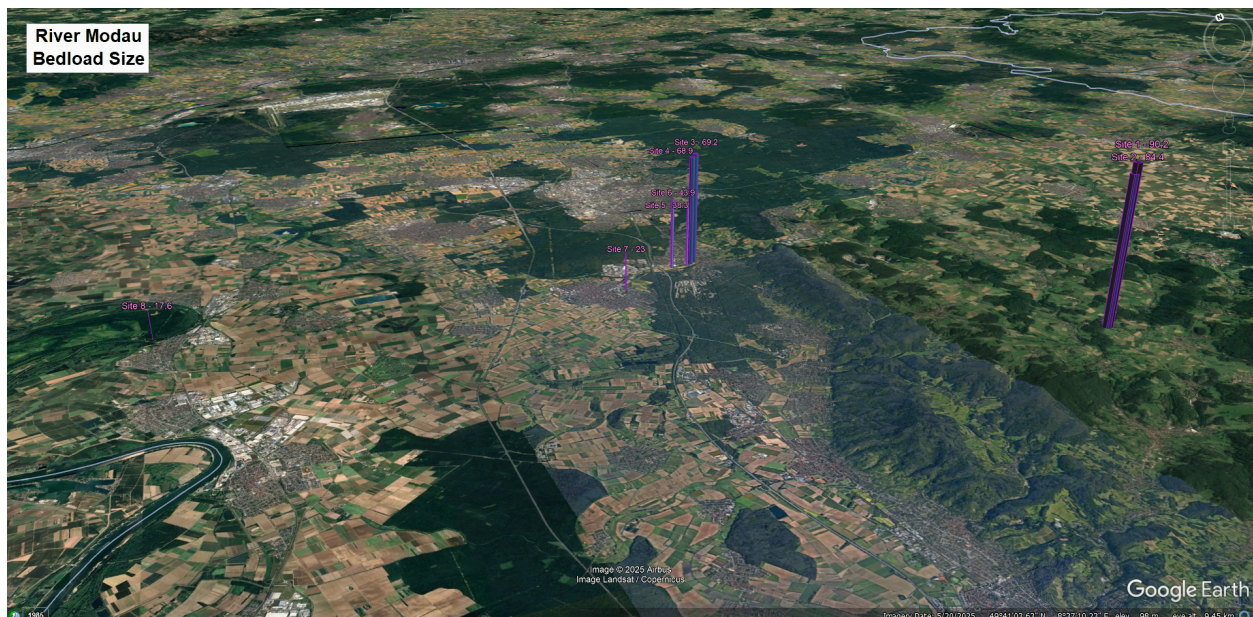


Figure 16: Bedload Size at Different Sites

5 Conclusion

5.1 Hypothesis 1

The obtained results vary completely from the expected values as although velocity is supposed to increase as channel efficiency increases, the results show the very opposite happening at all sites except for at Site 8, where the data corresponds to one another. This goes against Bradshaw's model where velocity increases as we move downstream.

5.2 Hypothesis 2

As mentioned in the hypothesis, increase in the river's discharge does correlate to the channel width and channel depth to a certain extent. When comparing the discharge with the channel width, one can notice a clear trend with the river discharge increasing as the width increased and the discharge decreasing as the channel width decreased. The only anomaly is at Site 8, where the channel width increases to 4.24 metres in comparison to Site 7 at 3.87 metres, however, the discharge decreases from 4.31 cubic metres per second. Channel width corresponds to Bradshaw's Model where it increases as we move downstream, however, channel depth doesn't support this. Discharge also supports Bradshaw's Model as it shows an increase moving downstream.

5.3 Hypothesis 3

Results obtained for the Bedload size completely support the hypothesis that they decrease in size as we move downstream, except for at Site 6, where it increases to 43.9mm from 38.3mm before dropping to 23.0mm at Site 7. More importantly, these results wholly support the Bradshaw Model.

6 Evaluation

There are many reasons for various measurements havng been totally inaccurate due to both natural and man-made causes. Velocity did not match up with Bradshaw's Model at almost all sites due to various hard- and soft-engineering being present. It was high at site 1 due to mattressing present underneath the bridge by site 1, then it slows down at site 2 due to various stones, branches, and other plant material due to its location in the forest. At Site 8, where the river is supposed to be at its fastest, it is at it's slowest, due to build up in sedimentation and silt.

7 Appendix

7.1 Raw Data

Sites	Recording 1	Recording 2	Recording 3	Recording 4
Site 1	0.59	0.53	0.60	0.57
Site 2	0.26	0.23	0.23	0.24
Site 3	0.93	1.16	1.24	1.11
Site 4	0.78	0.72	0.90	0.80
Site 5	0.12	0.24	0.21	0.19
Site 6	0.34	0.20	0.16	0.23
Site 7	0.28	0.18	0.40	0.29
Site 8	0.01	0.06	0.03	N/A

Table 1: Velocity Digital

Sites	Cross-sectional Area A (m ²)	Wetted Perimeter P (m)	Channel Efficiency $R = A/P$ (m)
Site 1	0.2929	2.03	0.14428571
Site 2	0.75001	2.96	0.25338176
Site 3	1.836051	6.91	0.26570926
Site 4	0.95328	5.02	0.18989641
Site 5	1.326024	6.75	0.196448
Site 6	1.167504	6.33	0.18443981
Site 7	0.73587	3.53	0.20846176
Site 8	1.1872	4.67	0.25421842

Table 2: Channel Efficiency Calculations per Site

Sites	Recording 1	Recording 2	Recording 3	Recording 4	Recording 5
Site 1	2.01	1.90	2.10	1.98	2.15
Site 2	3.00	3.02	2.95	2.80	3.05
Site 3	6.74	7.00	7.00	6.95	6.85
Site 4	5.12	5.00	4.95	5.09	4.94
Site 5	6.70	6.60	6.85	6.80	6.82
Site 6	6.48	6.20	6.29	6.36	6.30
Site 7	3.60	3.40	3.53	3.57	3.55
Site 8	4.70	4.62	4.78	4.60	4.65

Table 3: Wetted Perimeter

Sites	Recording 1	Recording 2	Recording 3
Site 1	2.00	2.37	1.69
Site 2	2.55	5.98	4.04
Site 3	4.40	4.39	4.42
Site 4	5.28	5.30	5.31
Site 5	7.01	7.01	7.03
Site 6	5.60	5.65	5.59
Site 7	3.83	3.90	3.98
Site 8	4.25	4.24	4.23

Table 4: Width

Sites	Recording 1	Recording 2	Recording 3
Site 1	0.141	0.138	0.155
Site 2	0.312	0.1375	0.084
Site 3	0.426	0.412	0.414
Site 4	0.188	0.182	0.170
Site 5	0.189	0.187	0.190
Site 6	0.213	0.211	0.201
Site 7	0.208	0.158	0.203
Site 8	0.273	0.284	0.283

Table 5: Depth

Sites	Cross-sectional Area A (m ²)	Velocity $\bar{v}$ (m/s)	Discharge $Q = A \times \bar{v}$ (m ³ /s)
Site 1	0.2929	0.5725	0.16768525
Site 2	0.75001	0.24	0.1800024
Site 3	1.836051	1.110	2.03801661
Site 4	0.95328	0.80	0.762624
Site 5	1.326024	0.19	0.25194456
Site 6	1.167504	0.2325	0.27144468
Site 7	0.73587	0.2875	0.21156263
Site 8	1.1872	0.03334	0.03958125

Table 6: Discharge Calculations per Site

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